



D.2 Electric and magnetic fields

Fields · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Coulomb's law states that point charges exert a force $F = kq_1q_2/r^2$ on each other ($k = 1/4\pi\epsilon_0$), producing an electric field $E = kQ/r^2$ around a charge Q , directed away from positive charges and towards negative ones; electric potential is $V = kQ/r$, and potential energy of a charge q is $E_p = qV$, with work done $W = q\Delta V$ moving a charge between two points. A uniform electric field between parallel plates has magnitude $E = V/d$ (V = potential difference, d = plate separation) and exerts a constant force $F = qE$ on a charge. Magnetic fields are produced by moving charges/currents; a straight current-carrying wire produces a field circling it (right-hand rule), and a moving charge or current in an external magnetic field B experiences a force $F = qvB \sin\theta$ (charge) or $F = BIL \sin\theta$ (wire), with direction given by the right-hand rule, always perpendicular to both v (or I) and B .

Worked example: follow the reasoning

Two point charges of $+4.0 \mu\text{C}$ and $-3.0 \mu\text{C}$ are separated by 0.30 m in a vacuum. Find the magnitude of the force between them. ($k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$)

$$F = kq_1q_2/r^2 = (8.99 \times 10^9)(4.0 \times 10^{-6})(3.0 \times 10^{-6})/(0.30)^2 \\ = (8.99 \times 10^9)(1.2 \times 10^{-11})/(0.090) = 0.1079/0.090 = 1.20 \text{ N}.$$

Answer: $F \approx 1.2 \text{ N}$ (attractive, since charges are opposite in sign).

Quick check · multiple choice

A charged particle moves at velocity v perpendicular to a uniform magnetic field B . In which direction does the magnetic force act on the particle?

- A Parallel to v
- B Parallel to B
- C Perpendicular to both v and B
- D Antiparallel to v

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A parallel plate capacitor has plates separated by 0.020 m with a potential difference of 400 V across them. [8]
 - (a) Calculate the electric field strength between the plates.
 - (b) An electron (charge $-1.6 \times 10^{-19} \text{ C}$, mass $9.11 \times 10^{-31} \text{ kg}$) is released from rest at the negative plate. Calculate the force on the electron and its acceleration.
 - (c) Calculate the speed of the electron just before it reaches the positive plate.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $E = V/d = 400/0.020 = 2.0 \times 10^4 \text{ V}\cdot\text{m}^{-1}$.

(b) $F = qE = (1.6 \times 10^{-19})(2.0 \times 10^4) = 3.2 \times 10^{-15} \text{ N}$. $a = F/m = 3.2 \times 10^{-15}/9.11 \times 10^{-31} = 3.51 \times 10^{15} \text{ m}\cdot\text{s}^{-2}$.

(c) Using energy: $qV = \frac{1}{2}mv^2 \rightarrow v = \sqrt{2qV/m} = \sqrt{[2(1.6 \times 10^{-19})(400)]/9.11 \times 10^{-31}} = \sqrt{[1.28 \times 10^{-16}/9.11 \times 10^{-31}]} = \sqrt{1.405 \times 10^{14}} = 1.19 \times 10^7 \text{ m}\cdot\text{s}^{-1}$.

Quick check answer

C. The magnetic force on a moving charge is $F = qv \times B$, which by the properties of the cross product is always perpendicular to both the velocity v and the field B .

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.